

Computational Time Machine (CTM)

A Measurement and Experimental Framework for Generational AI R&D Systems

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September 10, 2026

Technical Preprint v2.0

Current version DOI: 10.5281/zenodo.22689836

Prior public version: Gorlanov (2026), *Computational Time Machine: Accelerated AI Civilizations for Compressing Technological R&D Time*, Zenodo, DOI: 10.5281/zenodo.22687555.

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Status: Research proposal and falsifiable systems hypothesis; not peer reviewed.

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Core thesis. The meaningful “time machine” is not transport into the future; it is the ability to compute and validate technological futures faster than chronological civilization can reach them.

Abstract

AI systems are increasingly able to perform long-horizon research tasks, generate and test hypotheses, write and execute code, search algorithm spaces, and participate in closed-loop laboratory workflows. These developments motivate a systems question that is distinct from simply adding more parallel agents: can an artificial research ecosystem accumulate validated discoveries across successive generations such that later researchers inherit a measurably better research environment than earlier ones? This paper develops the **Computational Time Machine (CTM)** as a falsifiable experimental framework for studying that question.

CTM models technological research as a persistent, instrumented, multi-role R&D system in which candidate results are generated, independently verified, promoted into an inheritable research substrate, and made available to successor generations. Inherited artifacts may include proofs, algorithms, datasets, simulators, software tools, experimental protocols, model components, or physical instrumentation. The architecture is called **Civilization-in-the-Loop (CitL)** because cumulative progress depends on functional analogues of specialization, competition, cooperation, memory, verification, resource allocation, tool-building, and succession rather than on one monolithic agent.

The paper formalizes the **Technological Time Compression Ratio (TTCR)** as a domain-specific ratio of externally validated progress rates between a CTM system and a matched reference R&D system. It introduces supporting measures for causal inheritance depth, reuse, replication, physical closure, compute efficiency, and energy efficiency; proposes controlled experiments that isolate the value of generational inheritance from ordinary parallelism; derives practical limits from serial dependencies, verification latency, coordination overhead, physical experiment duration, compute, and energy; and specifies safety gates that prevent the research substrate, evaluators, or governance rules from being modified without independent authorization.

The central claim is deliberately empirical. CTM is successful only if matched experiments show reproducible, causally cumulative progress that exceeds flat agentic baselines after controlling for compute, tools, information, and experimental budgets. If inheritance fails to improve validated progress, if gains disappear under independent replication, or if coordination, physics, energy, or safety costs dominate, the strong time-compression hypothesis is falsified for that regime.

Keywords: automated scientific discovery; AI agents; recursive self-improvement; self-driving laboratories; research automation; open-ended search; technological forecasting; cumulative innovation; TTCR; Civilization-in-the-Loop.

1. Introduction

The unit of AI-assisted research is shifting. Early systems primarily answered questions or generated isolated pieces of code. More recent systems can execute multi-step research workflows, generate

hypotheses, run experiments, analyze results, draft papers, search algorithm spaces with automated evaluators, and interact with physical laboratories [2-10]. Frontier laboratories are also beginning to measure AI-agent effort as a material component of research labor. OpenAI reported in September 2026 that, by mid-August, its research organization was using approximately 3.1 agent-workdays of effort for each human workday, while people continued to set priorities and judge results [13]. OpenAI’s Recursive Self-Improvement (RSI) hiring materials explicitly describe a goal of building systems that accelerate and ultimately conduct high-quality research, including evaluations for research judgment, hypothesis generation and testing, and long-horizon experiment execution [14].

These developments establish a powerful first abstraction: **research labor can increasingly be instantiated computationally**. But technological progress is not equivalent to the number of concurrent researchers. Historical R&D is cumulative. New generations inherit mathematics, software, experimental apparatus, manufacturing methods, datasets, scientific instruments, organizations, and theories produced by preceding generations. A faster research civilization must therefore do more than run a large swarm against a fixed environment. It must be able to *change the environment in which the next generation researches*.

The Computational Time Machine (CTM) is a framework for testing whether that intergenerational structure can be engineered and measured. The proposal does not assume that technological time is literally interchangeable with calendar time. It does not claim that arbitrary centuries of social, biological, or physical development can be compressed. Instead, it asks a narrower and measurable question:

Under matched initial conditions and resources, can a verified, generational AI R&D ecosystem produce cumulative frontier progress at a higher validated rate than a flat agentic R&D system whose tools and capabilities remain fixed?

The key mechanism is **verified inheritance**. Research generation g produces candidate artifacts. Independent validators test correctness, novelty, reproducibility, transfer value, and safety. Only artifacts that cross pre-registered promotion thresholds become part of the research substrate available to generation $g+1$. A promoted artifact is not merely summarized in memory. It may alter the effective capability of the system: a faster solver, a better theorem tactic, a new simulator, a new experimental protocol, a superior compiler optimization, a calibrated world model, or a validated instrument can become a tool used by later generations.

This distinction matters because current automated-discovery systems already demonstrate many components of the loop. The AI Scientist and AI Scientist-v2 automate substantial parts of the machine-learning research cycle [2,3]. Google’s AI co-scientist uses a multi-agent architecture for generating and refining scientific hypotheses [4]. FunSearch and AlphaEvolve combine generative models with automated evaluators to discover verifiable programs and algorithms [5,6]. The Darwin Godel Machine explores self-improving coding agents through empirically validated code modification [7]. A-Lab and SAMPLE demonstrate closed-loop physical experimentation in materials science and protein engineering [8,9]. CTM builds on these directions but makes the *persistent, cross-generation research substrate itself* the object of study.

The contribution of this paper is therefore not a claim that automated science, evolutionary search, self-improving agents, or robotic laboratories are new individually. The proposed contribution is the combination of: (i) a generational R&D architecture in which verified discoveries can modify successor research capabilities; (ii) a controlled experimental design separating inheritance effects from parallel-agent effects; (iii) TCCR as an operational rate comparison rather than a rhetorical “years in days” claim; (iv) causal metrics that require later discoveries to depend on earlier machine-generated advances; (v) explicit compute, energy, physical-latency, and safety accounting; and (vi) a staged path from digital domains to bounded physical closure.

2. Related work and the systems gap

2.1 Automated scientific agents

The AI Scientist demonstrated an end-to-end framework that generates research ideas, writes code, executes experiments, produces figures and papers, and uses an automated reviewer [2]. AI Scientist-v2

extended this direction with progressive agentic tree search and removed reliance on human-authored code templates; one fully AI-generated manuscript exceeded the acceptance threshold at an ICLR workshop [3]. Google’s AI co-scientist similarly uses multiple agents to generate, debate, evolve, and rank hypotheses under scientist-specified objectives [4]. These systems show that substantial portions of the scientific workflow can be orchestrated by AI.

CTM asks a different longitudinal question. If one automated research cycle yields a useful artifact, does the *next generation* merely read the artifact, or does the artifact become part of an improved research environment whose effect can be measured? The distinction is between repeated autonomous projects and cumulative research infrastructure.

2.2 Verifiable search and evolutionary discovery

FunSearch pairs a language model with systematic evaluation and an evolving program database, producing new mathematical constructions and improved algorithms [5]. AlphaEvolve extends evaluator-driven evolutionary search to a wider range of algorithmic and scientific problems and has been used to optimize parts of computational infrastructure [6]. These systems illustrate a crucial CTM principle: generative creativity becomes more scientifically useful when coupled to hard evaluators.

CTM generalizes the evaluator loop from “candidate program -> score -> next candidate” to “candidate research artifact -> independent validation -> controlled promotion -> changed successor environment.” Evolutionary search may be one component of a CTM resource allocator, but the architecture permits multiple organizations, heterogeneous models, different experimental paradigms, adversarial replication teams, and physical evidence.

2.3 Self-improving agents

The Darwin Godel Machine (DGM) demonstrates open-ended empirical improvement of coding agents that modify their own code and retain an archive of successful variants [7]. Related self-improving coding-agent work shows that agents can autonomously edit their own scaffolding and improve benchmark performance. These results are directly relevant because they operationalize capability-changing artifacts and empirical validation.

CTM differs in scope and safety boundary. It does not require an agent to rewrite itself, nor does it define recursive self-improvement as the primary objective. A CTM generation may leave the foundation model unchanged while improving theorem provers, search procedures, data pipelines, experimental equipment, simulators, memory systems, or training curricula. Capability-changing artifacts are promoted by an external gate; evaluators and governance rules are not self-approved. This makes it possible to study cumulative acceleration without assuming unrestricted autonomous self-modification.

2.4 Self-driving laboratories

Self-driving laboratories combine algorithmic experiment selection with automated execution, measurement, and iterative learning [8-11]. A-Lab integrated computation, literature-derived synthesis knowledge, machine learning, and robotics to realize inorganic materials in a closed-loop workflow [8]. SAMPLE connected an intelligent agent to a fully automated protein-engineering laboratory and searched a small fraction of a large sequence landscape while identifying thermostable enzymes [9]. Reviews of self-driving laboratories emphasize both their acceleration potential and the importance of reproducibility, interoperability, autonomy definitions, and meaningful performance metrics [10,11].

These systems provide the physical-closure layer required by CTM. They also expose a hard limit: reality has latency. A chemical reaction, biological growth cycle, long-duration materials test, fabrication step, or astronomical observation cannot be accelerated simply by adding language-model tokens. CTM therefore treats physical experiments as privileged evidence and explicitly accounts for non-compressible physical time.

2.5 Frontier research acceleration and safety

OpenAI’s FrontierScience benchmark emphasizes that research-level scientific contribution requires more than fact recall and that open-ended research remains difficult even for capable models [12]. Its 2026 field reports describe rapidly increasing use of coding agents in research and scientific software development [13,16]. In September 2026, OpenAI also stated that fully autonomous recursive self-improvement is not occurring today and should not be pursued unless it can be done safely, while arguing for supervised automated AI researchers that can help improve subsequent systems [15].

CTM adopts this distinction. The initial experimental agenda is a **bounded, supervised research-acceleration system**, not a blueprint for uncontrolled recursive self-improvement. Successor capability changes are inspectable releases with explicit provenance, external evaluation, authorization, and rollback.

2.6 The systems gap

Existing systems cover many CTM components, but the following conjunction remains the target of the proposed experiments:

Dimension	Typical current focus	CTM experimental requirement
Time structure	One project or iterative search loop	Explicit research generations with successor inheritance
Capability state	Mostly fixed model/tool environment	Validated artifacts may change later tools and research infrastructure
Verification	Task evaluator, reviewer, or lab measurement	Independent, provenance-aware promotion gate with replication
Population	One agent, agent team, or evolutionary population	Multiple specialist and adversarial roles with resource allocation
Progress metric	Task score, paper score, optimization objective	Validated progress rate relative to a matched reference system
Causality	Better final output	Later advances must demonstrably depend on earlier inherited advances
Physical world	Domain specific	Simulation-to-physical escalation treated as a core architectural layer
Safety	System specific	Promotion, external action, and capability changes separately governed

3. Operational model: Civilization-in-the-Loop

3.1 Research generations

Let the research substrate available at generation g be

$$S_g = (K_g, T_g, M_g, D_g, X_g, R_g, G_g),$$

where K is validated knowledge, T is the toolchain, M is the permitted model set and agent scaffolding, D is data and provenance, X is simulation and experimental infrastructure, R is resource-allocation state, and G is governance and safety policy. A generation is a bounded research epoch that begins with a frozen view of S_g and produces a set of candidate artifacts A_g .

Artifacts may be scientific claims, proof objects, source code, datasets, model weights or adapters, workflows, simulator improvements, experimental protocols, hardware designs, calibration models, or instrument-control software. Each artifact carries a provenance record linking inputs, prompts, code, model versions, experiment logs, seeds, data lineage, compute use, and dependent artifacts.

A validator V evaluates candidate artifacts using domain-specific criteria. A promotion operator Π accepts only artifacts satisfying pre-registered thresholds for correctness, novelty or utility, reproducibility, provenance completeness, compatibility, and safety:

$$P_g = \Pi(A_g, V, G_g).$$

The next research substrate is constructed by a controlled inheritance operator H :

$$S_{g+1} = H(S_g, P_g),$$

subject to two important restrictions. First, H is reversible: promoted artifacts are versioned and can be rolled back. Second, the system cannot autonomously weaken the promotion rules, safety policy, or independent evaluator boundary. Proposed changes to those components require a separate governance path.

This process is called the **generational compiler**. The term does not imply that every artifact is software. It denotes the controlled conversion of validated research outputs into successor capabilities.

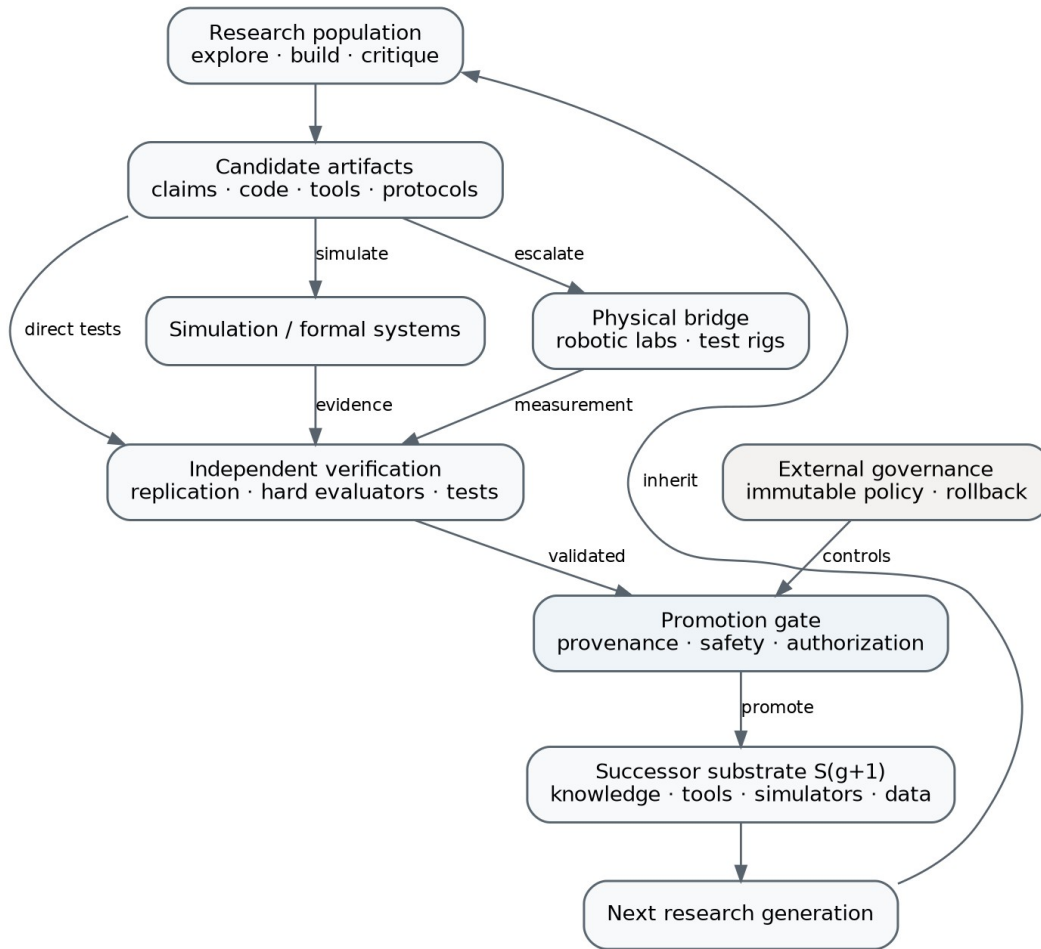


Figure 1. CTM generational research loop. Candidate work becomes successor capability only after independent verification and an externally governed promotion gate.

3.2 Functional research roles

A useful CitL implementation should avoid a homogeneous swarm. Correlated model errors and shared assumptions can make apparent consensus meaningless. The research population therefore contains functional roles with partially independent incentives and contexts:

- **Explorers** search broadly for hypotheses, designs, and unexpected opportunities.
- **Specialists** pursue high-depth work in mathematics, software, engineering, or domain science.
- **Builders** convert ideas into executable code, instruments, simulations, and testable protocols.
- **Adversarial critics** attempt to falsify claims and identify hidden assumptions.
- **Replication teams** independently reproduce results without access to private reasoning traces from the originating lineage.
- **Formal validators** use theorem provers, static analysis, model checkers, or exact evaluators where available.
- **Experimental operators** control bounded physical or simulated experiments.
- **Resource allocators** assign compute and experiment budgets to competing branches.

- **Archivists** maintain provenance graphs, dependency records, uncertainty, and supersession history.
- **Governance agents and humans** enforce domain boundaries, hazard rules, release gates, and rollback.

These roles need not map one-to-one to foundation-model instances. They can be implemented by separate prompts, models, fine-tunes, tool permissions, or organizations. The design objective is epistemic and operational diversity, not anthropomorphic simulation.

3.3 Branching technological trajectories

A single linear research lineage risks premature convergence. CTM therefore permits branches $b \in B_g$, each inheriting a controlled subset of the validated substrate and operating under different allocations, hypotheses, model mixtures, or constraints. Branches are evaluated as portfolios: successful stepping stones can be merged, while failed lineages remain valuable negative evidence.

A branch is not accepted because its terminal story sounds plausible. It must expose a dependency graph showing how each later capability was obtained. Branching therefore transforms technological forecasting from narrative extrapolation into **constructive trajectory search**: the system attempts to build a reachable chain of intermediate advances, each with evidence.

4. Measuring technological time compression

4.1 Why “years of progress” is not a primitive unit

Calendar years are an invalid universal research unit. One year in theorem proving, semiconductor fabrication, drug development, and astrophysics has different bottlenecks. A credible CTM measurement must therefore be local to a benchmark domain, externally scored, and compared with a reference R&D process under controlled conditions.

For domain D , define a pre-registered scoring function $s_D(a)$ for independently validated artifact a . The scoring function can be objective when the domain permits it: proof correctness, held-out benchmark improvement, measured physical performance, verified reduction in runtime or energy, or achievement of a hidden engineering target. Ambiguous artifacts can be scored by blinded expert panels, but such scores should be reported separately from hard-evaluator scores.

The cumulative validated progress of system X during horizon T is

$$P_D(X, T) = \sum_{a \in A_{X,T}^{\text{validated}}} s_D(a).$$

No credit is assigned to outputs that fail independent verification, duplicate known results under the experiment’s knowledge cutoff, leak hidden test information, or cannot be reproduced according to the pre-registered protocol.

4.2 Technological Time Compression Ratio

The principal metric is the ratio of validated progress rates:

$$TTCR_D = \frac{P_D(CTM, T_C)/T_C}{P_D(REF, T_R)/T_R}.$$

REF is a matched reference system, not an imagined historical civilization. It may be a flat multi-agent research system, a human+AI team, or another automated research architecture. The initial substrate, accessible information, model family, and resource budgets must be specified.

When both systems run for equal wall-clock time T , the expression reduces to a progress ratio. When both are required to reach a fixed target progress P^i , $TTCR$ can be measured as a time ratio T_R/T_C . Repeated runs produce confidence intervals rather than a single promotional number.

A task-level work metric can also be reported when credible human baselines exist:

$$TTCR_{work} = \frac{\text{validated reference human-hours}}{\text{CTM wall-clock hours}},$$

but this should not be extrapolated to open-ended technological history.

4.3 Causal inheritance depth

High throughput alone does not demonstrate a computational time machine. A million independent discoveries made in parallel may have large economic value but zero intergenerational depth. CTM therefore requires a dependency directed acyclic graph (DAG) over promoted artifacts. Define **Causal Inheritance Depth (CID)** as the maximum validated dependency path length from an artifact produced during the experiment to a later promoted artifact that materially depends on it.

A CTM result with TTCR much greater than 1 but CID = 1 is primarily a parallel research accelerator. A stronger demonstration requires both high validated rate and multi-generation causal dependence.

4.4 Supporting metrics

TTCR should be accompanied by metrics that diagnose *why* acceleration occurs:

Metric	Purpose
Independent replication rate	Fraction of promoted claims reproduced by a blinded lineage
Promotion precision	Fraction of promoted artifacts surviving later audit or re-test
Inheritance uplift	Improvement in successor performance attributable to inherited artifacts versus a no-inheritance counterfactual
Validated reuse rate	Fraction of later progress that uses promoted machine-generated artifacts
Causal inheritance depth	Depth of cumulative dependency, not just parallel output count
Diversity retention	Whether multiple viable research branches persist rather than collapsing to one correlated lineage
Physical closure rate	Fraction of simulation-dependent claims that survive required real-world testing
Progress per joule	Validated progress divided by total computational and experimental energy
Progress per dollar	Validated progress divided by total cost under a declared accounting boundary
Human oversight load	Human review/approval hours per unit of validated progress
Safety intervention rate	Frequency and severity of blocked or rolled-back transitions

The important methodological principle is that a single headline TTCR must not hide falling replication, exploding energy use, or a growing human-validation bottleneck.

5. Experimental design

5.1 Primary hypothesis

The primary hypothesis is:

H1: *Under matched initial substrate, model family, information access, total compute, and experimental budget, a verified generational system produces a higher rate of independently validated cumulative progress than a flat agentic research system with a fixed tool environment.*

The null hypothesis is that any apparent advantage is explainable by ordinary parallelism, additional memory, evaluator leakage, more compute, or statistical noise.

5.2 Experimental arms

The minimum controlled experiment should compare the following arms:

Arm	Shared memory	Tool environment can change	Independent promotion gate	Branching
A. Flat agentic baseline	Yes	No	Yes	Optional

Arm	Shared memory	Tool environment can change	Independent promotion gate	Branching
B. CTM generational	Yes	Yes, after validation	Yes	Yes
C. CTM no-tool-inheritance ablation	Yes	No	Yes	Yes
D. CTM single-lineage ablation	Yes	Yes	Yes	No

All arms receive the same starting models, knowledge snapshot, benchmark descriptions, network policy, and total resource envelope. Arm A is deliberately strong: it may share research notes and previous results. The variable of interest is not memory alone; it is whether validated outputs can become *new capabilities of the environment*.

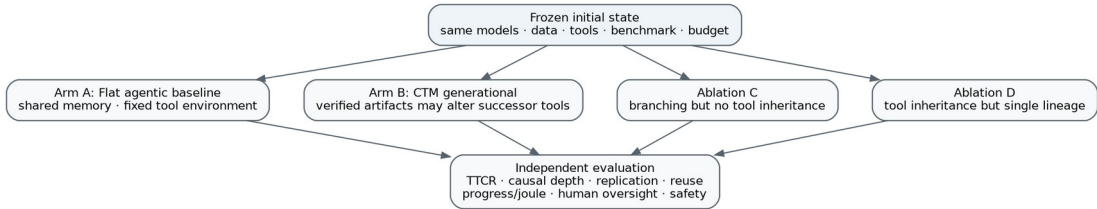


Figure 2. Controlled evaluation separates ordinary parallel-agent productivity from the additional effect of verified generational inheritance.

5.3 Domain selection

The first demonstration should prioritize domains where novelty and correctness can be checked cheaply and independently.

Track 1 - digital, hard-verification domains. Candidate domains include theorem proving, compiler optimization, algorithm design, formal methods, constrained chip placement/routing, software performance, and selected computational-science tasks. The ideal task contains a large search space, hard held-out evaluators, and opportunities for tool-building that later generations can reuse.

Track 2 - simulated engineering domains. Digital twins or high-fidelity simulators can test control systems, circuit designs, scheduling policies, or bounded mechanical designs. Final evaluators should be hidden from research agents to reduce reward hacking. Simulator uncertainty must be explicitly modeled.

Track 3 - bounded physical domains. After digital validity is established, connect the architecture to self-driving laboratory or automated test infrastructure. Suitable early targets are those with relatively short experiment cycles, strong instrumentation, manageable hazards, and objective endpoints. Examples include optical alignment, electronics characterization, selected materials workflows, and chemistry restricted to approved reagents and reaction classes.

5.4 A 30-day first demonstration

A defensible first milestone can be run as a 30-day research campaign in one or two high-verifiability digital domains. The following protocol is intentionally conservative:

1. Freeze a knowledge snapshot, starting toolchain, model versions, and benchmark suite.
2. Pre-register progress scoring, validation rules, safety policy, and compute/energy accounting.
3. Allocate identical total compute to baseline and CTM arms, with multiple independent seeds.
4. Divide CTM operation into explicit generations (for example, fixed compute epochs rather than arbitrary calendar periods).
5. Require independent reproduction before promotion of high-impact artifacts.
6. Maintain hidden final evaluations and contamination checks.
7. Record a dependency DAG for every promoted artifact and every later result that uses it.
8. Run no-inheritance and no-branching ablations.
9. Re-test a sample of promoted artifacts after several generations to estimate promotion precision.
10. Report TTCR, CID, replication, reuse, energy, cost, oversight load, and all failures.

The experiment should not be declared successful because the system produced more papers, tokens, hypotheses, or benchmark submissions. A stronger result is a sequence such as: generation 1 discovers a verified optimization; generation 2 incorporates it into a tool; generation 3 uses the improved tool to reach a result that the fixed-tool baseline does not reach under equal resources; generation 4 independently reproduces and extends that result.

5.5 Statistical and reproducibility requirements

Agentic research is stochastic, and the tails may dominate outcomes. Single-run demonstrations are therefore weak evidence. The evaluation should use multiple randomized seeds, bootstrap or other appropriate confidence intervals, and pre-specified stopping rules. Where budgets prevent many full 30-day runs, smaller proxy environments should be used for power analysis and ablations before a large campaign.

Artifacts, evaluators, prompts, model versions, random seeds, environment images, and scoring code should be versioned. Independent teams should be able to replay at least a representative subset of promoted results. Claims of novelty should be checked against the frozen knowledge snapshot and, after the experiment, against the live literature to distinguish experiment-local novelty from world novelty.

6. Compute, energy, coordination, and physical limits

6.1 An Amdahl-style limit for research acceleration

Research acceleration contains both parallelizable and serial components. A simplified wall-clock model is

$$T_{CTM} = T_{serial} + \frac{T_{parallel}}{N_{eff}} + T_{coord} + T_{verify} + T_{physical} + T_{governance},$$

where N_{eff} is effective rather than nominal parallelism. Coordination overhead can grow with population size; verification often becomes a bottleneck as candidate volume rises; physical experiments may have fixed durations; and high-impact transitions may require human authorization.

This is analogous to Amdahl’s law [18]: unlimited parallel agents cannot eliminate serial dependencies. The purpose of CTM is partly to discover whether inherited tools can *shrink the serial terms themselves*. For example, a better simulator may reduce the number of physical trials, a better proof tactic may shorten verification, and improved instrumentation may accelerate measurement. If generations merely add more agents without changing these bottlenecks, TTCR should saturate.

6.2 Coordination as an endogenous research problem

At high concurrency, research coordination is not free. Duplicate work, conflicting assumptions, context transfer, artifact compatibility, and evaluator congestion can erase gains. CTM therefore treats coordination mechanisms as inheritable artifacts subject to the same validation rules. A better scheduler, artifact retrieval system, branch-merging protocol, or scientific memory architecture can be promoted if it improves externally measured progress under matched resources.

This creates a useful distinction between **population scaling** and **civilizational scaling**. Population scaling increases the number of researchers. Civilizational scaling improves the institutions and tools that allow a larger population to produce coherent cumulative progress.

6.3 Energy accounting

Compute is not an abstract unlimited input. The total energy boundary should include, where measurable,

$$E_{total} = E_{compute} + E_{storage/network} + E_{cooling/overhead} + E_{lab} + E_{fabrication}.$$

For data-center compute, PUE or an equivalent facility overhead should be reported rather than quoting accelerator power alone. Physical experiments should include major robotic, thermal, vacuum, or fabrication energy where relevant. The primary efficiency metric is

$$\eta_E = \frac{P_D}{E_{total}}.$$

A CTM system that doubles validated progress while increasing energy by an order of magnitude may be strategically unattractive even if its wall-clock TTCR is greater than one. Conversely, inherited algorithmic improvements can raise both time compression and progress per joule. Energy efficiency should therefore be part of the selection pressure, not an afterthought.

Long-horizon implementations may benefit from improved accelerators, cooling, power systems, or unconventional infrastructure. CTM does not assume any particular energy technology. Infrastructure innovation can itself become an R&D target, but the experimental metric must count the real resource cost rather than treating future compute as free.

6.4 Physics as a hard boundary

Some processes cannot be accelerated computationally because the measured phenomenon itself has duration. Long-term toxicity, biological aging, corrosion, climate response, and certain reliability phenomena are examples. CTM can use parallel cohorts, accelerated stress tests, surrogate models, and simulation, but these are epistemic approximations that require calibration. The architecture must represent uncertainty and explicitly escalate claims to real experiments when the simulator's validity envelope is exceeded.

A strong CTM result therefore does not imply unlimited time compression. It maps a **compression frontier**: the maximum defensible TTCR attainable in a domain at a given error tolerance, physical-validation requirement, resource budget, and safety constraint.

7. Verification, provenance, and epistemic control

7.1 Independent validation before inheritance

The most dangerous failure mode in generational systems is not a single hallucination; it is a hallucination that becomes infrastructure. Once an incorrect claim is inherited, later generations may build apparently coherent results on top of it. CTM therefore separates generation from promotion.

High-value artifacts should be evaluated through multiple independent channels when possible: exact tests, theorem proving, blinded replication, static analysis, adversarial review, benchmark holdouts, simulator cross-checks, and physical measurement. Validators should not automatically share the originating agents' scratchpads or private rationales because shared context can reproduce the same mistake.

7.2 Provenance graph

Every promoted artifact should have an immutable provenance graph recording:

- originating lineage and model/tool versions;
- input datasets and knowledge snapshot;
- code, parameters, seeds, and environment hashes;
- experimental raw data and transformations;
- claims and uncertainty boundaries;
- independent replication results;
- parent artifacts and downstream dependents;
- safety review and authorization events;
- supersession, rollback, or invalidation history.

This graph supports both science and safety. Scientifically, it allows causal analysis of cumulative progress. Operationally, it allows a compromised or invalid artifact to be traced and rolled back along its dependency descendants.

7.3 Preventing evaluator capture

A generational system has a strong incentive to optimize its evaluators. That is desirable when it finds genuinely better solutions and dangerous when it finds shortcuts, benchmark leakage, or ways to modify the score. Evaluator code, hidden tests, and promotion policy should therefore sit outside the ordinary writable research substrate. Proposed evaluator improvements can be tested, but changing the judge of success requires an independent governance path and retrospective comparison against old evaluators.

8. Safety architecture

The CTM research agenda should be separated from unrestricted recursive self-improvement. OpenAI's September 2026 policy statement explicitly distinguishes current research acceleration from fully autonomous RSI and argues that the latter should not be pursued without adequate safety [15]. CTM is designed so the first experiments can study cumulative research acceleration while preserving meaningful human control.

The minimum safety architecture is:

1. **Domain whitelisting.** Research is restricted to pre-approved domains, datasets, tools, and physical experiment classes.
2. **Least-privilege agents.** Research agents receive only the permissions needed for their role; external network actions and account operations are separately gated.
3. **Immutable governance boundary.** Agents cannot unilaterally relax safety rules, evaluator separation, logging, or promotion thresholds.
4. **Capability-change review.** A promoted artifact that materially increases autonomy, cyber capability, model-training capability, or external-action scope receives an elevated review path.
5. **Independent safety budget.** Safety, security, interpretability, and red-team branches have protected resources and can block promotion.
6. **Sandboxed execution.** Candidate software and models execute in isolated environments with controlled data egress and reproducible images.
7. **Physical hazard controls.** Robotic experiments use approved reagent/material lists, bounded energy and temperature envelopes, interlocks, and human-authorized escalation.
8. **Dual-use screening.** High-risk biological, chemical, cyber, weapons, or surveillance outputs are excluded or routed to appropriate controlled evaluation rather than inherited into the general substrate.
9. **Rollback and quarantine.** Every generation has known-good checkpoints; suspicious artifacts and their descendants can be quarantined.
10. **Human authorization for irreversible transitions.** The objective is supervised cumulative acceleration, not removal of human authority over system scope.

A useful safety success criterion is *oversight scalability*: as research throughput rises, the number of decisions requiring direct human review should not grow linearly with candidate volume. Automated validation can absorb routine cases, while human review remains concentrated on high-impact transitions. This is itself measurable through human oversight load.

9. Failure modes and falsification criteria

CTM is a research hypothesis, not a promised outcome. The following results would weaken or falsify important versions of the thesis.

No inheritance advantage. After controlling for compute and memory, the generational arm performs no better than the flat agentic baseline. This would show that current tool-building opportunities or agent capabilities are insufficient to produce compounding research gains.

Shallow causality. TTCR is high but CID remains near one. The system is an effective parallel accelerator, not yet a generational time-compression system.

Verification collapse. Candidate output increases faster than independent validation, causing promotion precision or replication rates to fall. Apparent acceleration is then epistemically invalid.

Coordination saturation. Additional agents or branches create enough duplication and communication overhead that effective progress per unit compute declines.

Physical bottleneck dominance. Digital exploration becomes cheap, but real-world verification remains the rate-limiting step. CTM may still improve experiment selection, but domain-level TTCR saturates near the physical closure rate.

Energy or cost inversion. Wall-clock progress rises while progress per joule or per dollar falls to an unacceptable level. This would constrain practical scale even if the technical hypothesis holds.

Reward hacking or evaluator capture. The system raises benchmark scores without producing transferable external improvements. Hidden tests, independent replication, and physical validation are intended to expose this failure.

Correlated epistemic failure. Multiple nominally independent agents reproduce the same false assumption because they share model training, prompts, or contaminated evidence. Model/tool diversity and blinded replication are necessary countermeasures.

Unsafe capability pressure. The fastest branches repeatedly seek to weaken controls or require unacceptable external capabilities. In that regime, safety constraints become a legitimate limit on TTCR rather than friction to be optimized away.

10. Staged experimental program

Stage I - Instrumented digital CTM

Build the minimum generational substrate in hard-verification software domains. Required components are an artifact registry, provenance DAG, isolated research branches, independent validators, a resource allocator, a promotion gate, and reproducible environment snapshots. The objective is to demonstrate that the experimental methodology works before pursuing open-ended science.

Exit criterion: a statistically credible inheritance uplift over a strong flat-agent baseline under matched compute, with replication and contamination controls intact.

Stage II - Multi-domain tool transfer

Introduce several digital research domains and test whether an artifact created in one domain improves another. Examples include a general experiment scheduler, theorem-search heuristic, retrieval system, or compiler optimization. Cross-domain transfer is important because historical technological acceleration often comes from general-purpose tools rather than isolated discoveries.

Exit criterion: validated cross-domain reuse with measurable improvement in progress rate and no loss of verification quality.

Stage III - Branching technological trajectory search

Run multiple research organizations with different assumptions and allocations. Require constructive dependency chains rather than narrative forecasts. Test portfolio allocation, branch merging, negative-result reuse, and diversity preservation.

Exit criterion: branches discover distinct validated stepping stones and later generations combine them into results not reached by linear baselines within the same budget.

Stage IV - Simulation-to-physical closure

Connect selected branches to bounded robotic or automated test infrastructure. Treat simulator uncertainty as explicit state and allow physical evidence to override inherited models.

Exit criterion: a meaningful fraction of simulation-derived improvements survives independent physical testing, with measured end-to-end TTCR including laboratory latency and energy.

Stage V - Research-infrastructure co-evolution

Allow validated improvements to research tooling, coordination, experiment automation, and model scaffolding to enter the substrate. High-impact model or autonomy changes remain separately governed.

Exit criterion: sustained improvement in validated progress per unit compute or energy across multiple generations, not merely increased raw spending.

11. What would constitute a defensible “time-compression” demonstration?

A strong demonstration should satisfy five conditions simultaneously.

First, it produces **frontier-relevant validated outputs**, not merely more text or rediscovery. Second, later outputs have **causal dependence** on machine-generated artifacts from earlier generations. Third, the improvement survives **independent replication or hard evaluation**. Fourth, the comparison uses a **matched reference system** and reports compute, energy, experiment, and human-oversight budgets. Fifth, the result is **reproducible across runs** or robust enough that uncertainty is quantified.

A possible milestone is:

In a pre-registered 30-day digital R&D experiment, a generational CTM system achieves a TTCR significantly greater than 1 relative to a flat agentic baseline under equal total compute; produces a multi-generation validated dependency chain; maintains predefined replication and promotion-precision thresholds; and demonstrates positive inheritance uplift after ablations for shared memory, branching, and evaluator access.

Only after such a result should stronger language about “years of research compressed into days” be calibrated. If a domain has an independently measured human or institutional progress baseline, the observed validated progress rate can be converted into an *equivalent reference time* within that domain. That equivalence must be reported with uncertainty and never generalized automatically to civilization-wide progress.

12. Discussion

12.1 Why the civilization metaphor is useful

The word “civilization” can sound grandiose if interpreted socially. In CTM it has a narrower systems meaning. Cumulative innovation requires more than intelligence. It requires durable memory, division of labor, institutions for trust and criticism, resource allocation, tool production, handoff between generations, and mechanisms for preserving useful discoveries while rejecting false ones. These are properties of a research civilization even if every participant is software.

This framing also clarifies why simply scaling agent count may underperform. A million researchers without reliable literature, instruments, replication, incentives, and coordination can generate enormous noise. The main object of optimization is therefore not only the researcher; it is the **research system**.

12.2 The strongest version of the hypothesis

The strongest CTM hypothesis is not merely that AI makes research faster. That is already increasingly plausible and measurable. The stronger hypothesis is that generational inheritance creates

compounding research capacity: validated outputs improve the production function for later outputs, which in turn improves the next generation's tools.

Compounding should not be assumed. It can saturate because easy optimizations are exhausted, evaluators become costly, physical reality dominates, or coordination overhead rises. The experimental program is designed to locate those transition points. A negative result would still be scientifically useful because it would identify which bottleneck prevents research labor from turning into technological time compression.

12.3 Alternative futures as experimental objects

Once constructive inheritance works, CTM can be used to explore alternative technological trajectories. Instead of asking a language model to predict a distant year, the system can attempt to *earn* each step through validated intermediate work. Different branches may prioritize energy efficiency, new algorithms, experimental instrumentation, alignment, or hardware. The resulting terminal states are not forecasts in the statistical sense; they are candidate reachable futures supported by explicit dependency chains.

This could make technology strategy more rigorous. A high-value endpoint that requires an implausible or unvalidated intermediate step would be rejected early. A less glamorous path with a strong chain of experimentally supported stepping stones could receive more resources.

12.4 Compute infrastructure and the energy frontier

At large scale, accelerator efficiency and power availability may determine the feasible research population. This makes hardware, datacenter systems, algorithms, and energy technologies part of the same feedback loop. A CTM system should favor improvements that reduce the energy cost of future research. In principle, the research civilization can spend resources on making its own computational substrate more efficient, provided such changes are validated and safely promoted.

The framework remains agnostic about whether future infrastructure is terrestrial, distributed, or eventually off-world. Those are engineering trajectories to evaluate, not assumptions required for the model. The invariant is resource accounting: progress should be measured against real energy, cost, latency, and material inputs.

13. Conclusion

The Computational Time Machine is a proposal to turn a provocative metaphor into a measurable systems experiment. The relevant question is not whether an AI can imagine the technology of 2150. It is whether an artificial R&D ecosystem can construct and validate a chain of technological advances faster than a matched reference research process, while later generations inherit the verified tools and discoveries of earlier ones.

The framework therefore places four requirements at the center: **verification before inheritance, causal cumulative progress, matched experimental comparison, and explicit physical/resource/safety accounting**. TTCR measures validated progress rate rather than tokens or agent-hours. Causal inheritance depth distinguishes genuine generational accumulation from parallel productivity. Self-driving laboratories provide a route to reality when simulation is insufficient. Safety gates keep changes to models, tools, evaluators, and external capabilities under controlled promotion rather than unconstrained self-modification.

If generational inheritance produces sustained gains under these constraints, technological progress becomes partially more computational and less calendar-bound. If it does not, CTM still provides a useful diagnostic framework for identifying the limiting factor: intelligence, tool transfer, coordination, verification, physics, energy, economics, or governance. In either case, the decisive step is empirical: build bounded research civilizations, measure them, and require the future to be earned by validated intermediate discoveries.

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Version note. This technical preprint is version 2.0 of the Computational Time Machine research line. Version-specific DOI: 10.5281/zenodo.22689836. It expands the public v1.0 concept deposited on Zenodo under DOI 10.5281/zenodo.22687555. The v1.0 DOI, files, and GitHub release remain unchanged; substantive future revisions should receive a new version number and version-specific DOI.